2. Zooming In: Machine Learning, Deep Learning, and Generative AI

How machines learn from data, and the three big families

Week 2 · CE/ARCE 2XX AI Literacy for Civil and Architectural Engineering



COLLEGE OF ENGINEERING
Civil & Architectural Engineering & Mechanics

Agenda

Today's plan

1. Quiz on last week (5–8 min)
2. From rules to learning: what a model is
3. Three ways machines learn — with CE and ARCE examples
4. Deep learning, why now, and generative AI
5. Data is a design material
6. Lab day: working with data; Homework 1

Recap

Last week in 60 seconds

1 No single definition

Use two properties as a lens: **autonomy** and **adaptivity**. AI is a matter of degree.

2 Talk precisely

Say “an AI method” or “an AI system” — and say what it learned from.

3 The nested map

AI ⊃ machine learning ⊃ deep learning ⊃ generative AI, plus search, logic, optimization, robotics.

4 AI literacy

Understand it, use it, **verify it**, and use it responsibly.

Learning objectives

By the end of today, you can…

- ✓ Contrast traditional programming (rules + data → answers) with machine learning (data + answers → rules)
- ✓ Identify the **features X** and the **target y** in an engineering dataset
- ✓ Tell supervised (regression, classification), unsupervised, and reinforcement learning apart
- ✓ Explain what makes deep learning and generative AI different, and why they took off now
- ✓ Judge whether a dataset is good enough — quality, quantity, representativeness

01

From rules to learning

What changes when a computer learns from examples

From rules to learning

Machine learning flips traditional programming

Traditional programming



e.g., Beam formula $M = qL^2/8$ (q = load, kN/m) + span and load → maximum moment

Machine learning



e.g., 1,030 mixes + their measured strengths → a strength-prediction model

From rules to learning

Write the rules — or learn them from data?

Write the rules when…

- The physics or the code provision is known (beam moment, Manning's equation)
- You must explain and certify every step
- There are few examples, or none

vs

Learn from data when…

- The relationship is too complex to write down (strength from 8 ingredients and age)
- You have many examples with known answers
- Patterns are hard to describe in words (what a crack “looks like”)

Most real engineering tools combine both: physics and rules, plus learned parts

From rules to learning

Rules or learning? You decide

3 min

Think · Pair · Share

- For each task: **write the rules** or **learn from data**? Why?
- **1.** Check that a steel beam's bending stress is below the allowable value
- **2.** Estimate the energy use of a new design from 10,000 past simulations
- **3.** Find cracks in 5,000 drone photos of a parking garage
- **4.** Compute peak stormwater runoff with the rational method, Q = CiA

Ask yourself: Do we know the rule? Can we write it down? Do we have many examples with known answers?

A model is a function that turns inputs into a prediction

Model

A **model** maps inputs to an output: **$\hat{y} = f(x)$**. In machine learning, the numbers inside f (its **parameters**) are learned from data, not written by hand.

- **x** = inputs (features) · **$\hat{y}$** ("y-hat") = the prediction · **y** = the true answer
- Example: f(mix proportions, age) → predicted strength in MPa
- "All models are wrong, but some are useful" — statistician George Box

From rules to learning

Features X go in; the target y comes out

Features X — 8 inputs (kg/m³, except age)								Target y
Cement	Slag	Fly ash	Water	Super-plast.	Coarse agg.	Fine agg.	Age (days)	Strength (MPa)
540	0	0	162	2.5	1040	676	28	**79.99**
540	0	0	162	2.5	1055	676	28	**61.89**
332.5	142.5	0	228	0	932	594	270	**40.27**
332.5	142.5	0	228	0	932	594	365	**41.05**
198.6	132.4	0	192	0	978.4	825.5	360	**44.30**

… 1,030 rows in total (m = 1,030 examples)

One row = one **example** · columns we know = **features** · the column we predict = **target** (or **label**)

02

Three ways machines learn

Supervised, unsupervised, and reinforcement learning

Three ways machines learn

The difference is the feedback the learner gets

LEARN FROM ANSWERS

Supervised

- Examples come **with** the right answer (label)
- Learns to predict y from X
- CE: strength from mix; crack or no crack

FIND STRUCTURE

Unsupervised

- Examples come **without** answers
- Finds groups and unusual cases
- ARCE: group buildings by energy use

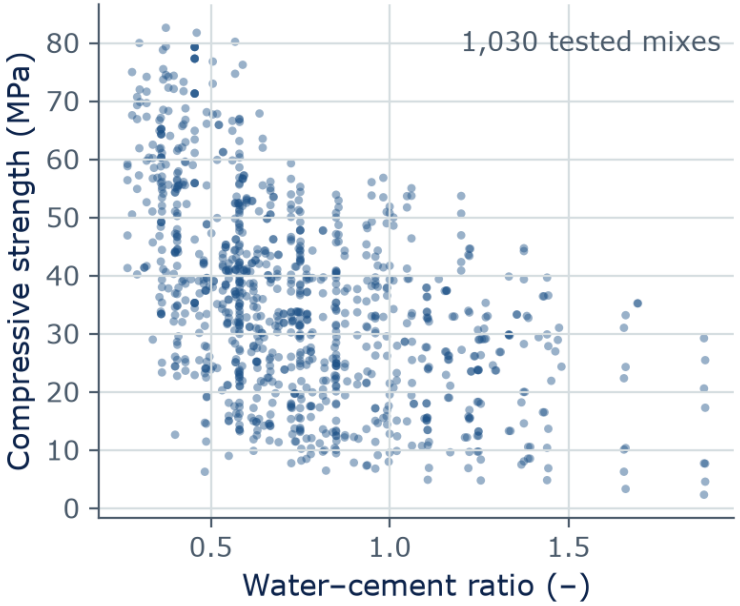
LEARN BY TRIAL

Reinforcement

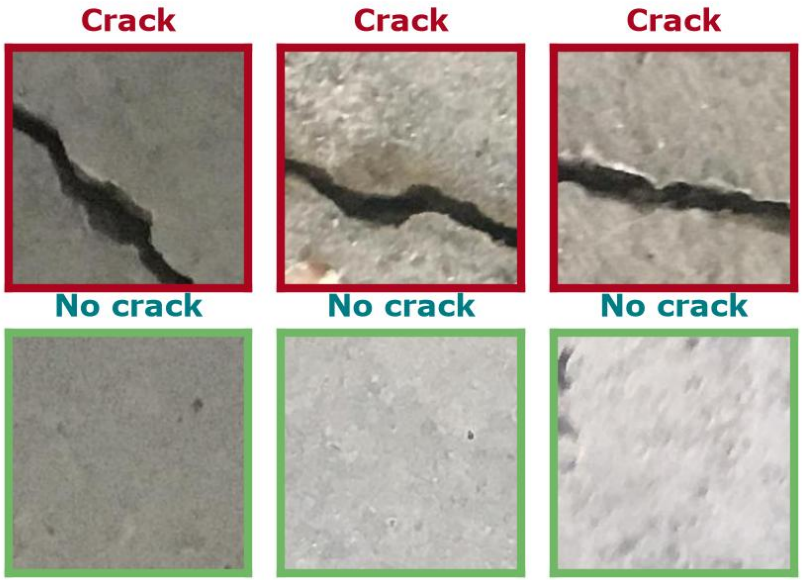
- Acts, then gets a **reward** or penalty
- Learns a strategy over time
- CE: traffic-signal timing; HVAC control

Supervised learning: regression predicts numbers, classification predicts categories

Regression: the output is a number

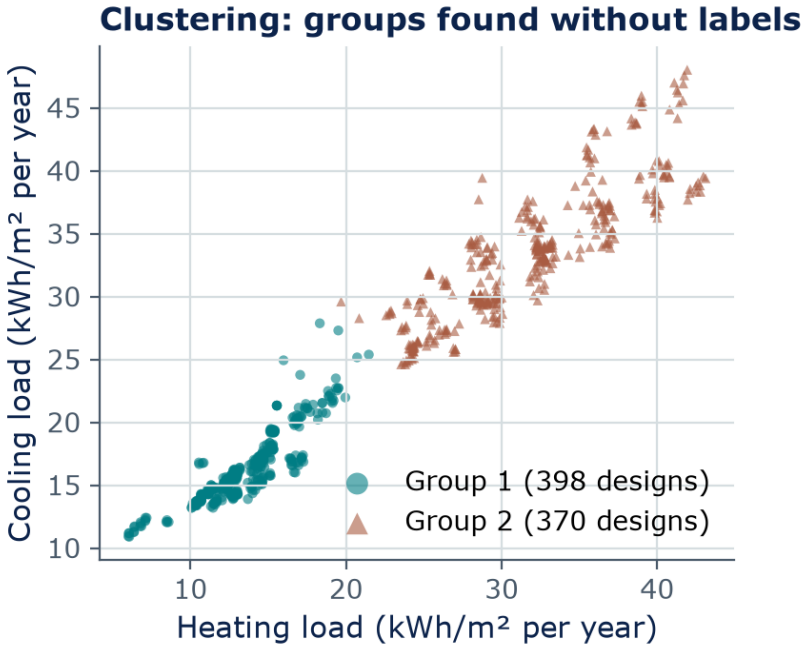


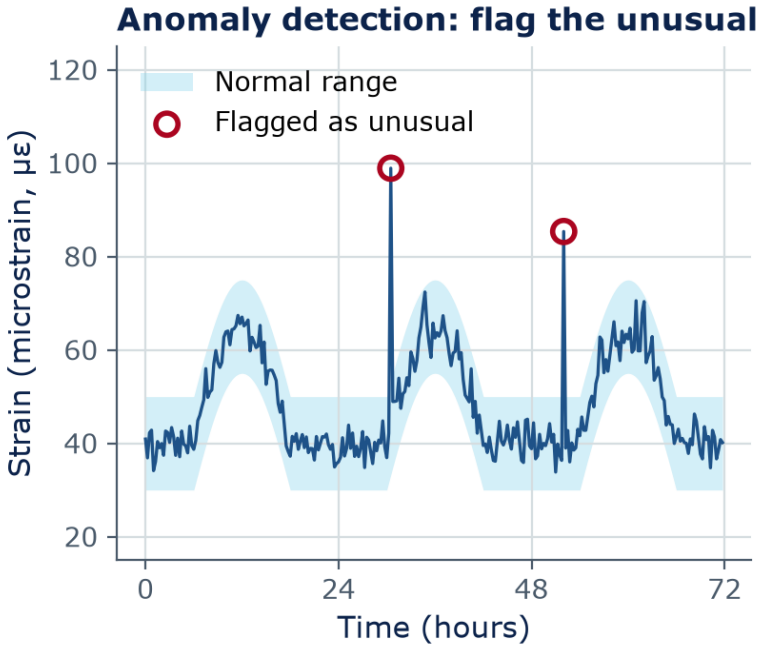
Classification: the output is a category



Data: Yeh (1998) concrete dataset; Özgenel (2019) crack images, CC BY 4.0 — both in the course data folder

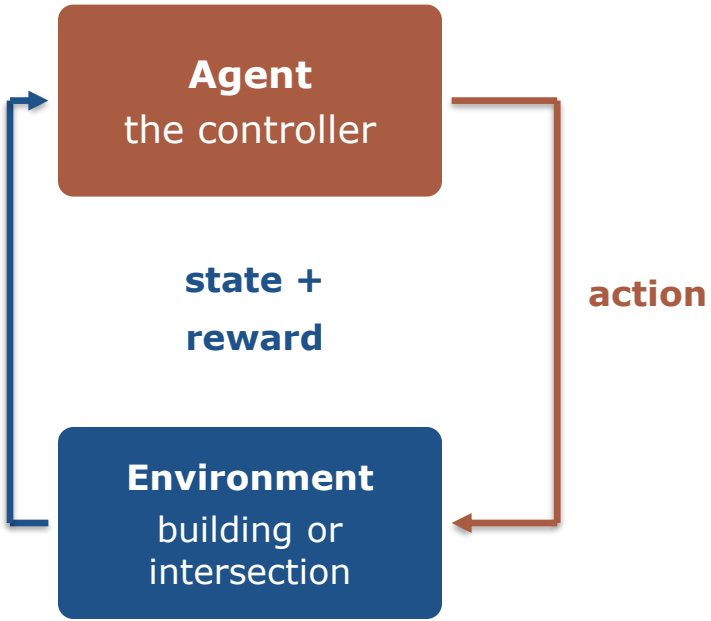
Unsupervised learning finds structure nobody labeled





Left: Tsanas & Xifara (2012) building energy data, k-means with 2 groups. Right: simulated signal, for illustration only.

Reinforcement learning: act, get a reward, improve



HVAC control (ARCE)

- **State:** room temperatures, occupancy, weather, energy price
- **Action:** heating and cooling setpoints
- **Reward:** low energy cost, comfortable people

Traffic-signal timing (CE)

- **State:** queue length on each approach
- **Action:** which phase gets green, and for how long
- **Reward:** less total delay

Side by side: three ways to learn

	Supervised	Unsupervised	Reinforcement
Data	Examples with answers (X and y)	Examples without answers (X only)	Experience: states, actions, rewards
Learns	A function: X → y	Groups, patterns, the unusual	A strategy: what to do when
Typical output	A number or a category	Group labels, anomaly scores	Actions
CE/ARCE example	Concrete strength; crack detection	Building energy groups; sensor anomalies	Signal timing; HVAC control
In this course	Weeks 3–6, most labs	Mentioned; HW1	Concept only; Week 10 agents

Concept check

Which kind of learning is this?

Traffic engineers have one year of hourly counts from 400 road sensors — no labels. They want to discover groups of sensors with similar daily patterns. This is...

A Supervised learning — regression

B Supervised learning — classification

C Unsupervised learning — clustering

D Reinforcement learning

Concept check

Which kind of learning is this? — answer

Traffic engineers have one year of hourly counts from 400 road sensors — no labels. They want to discover groups of sensors with similar daily patterns. This is...

A. Supervised learning — regression

B. Supervised learning — classification

C. Unsupervised learning — clustering

D. Reinforcement learning

No answers (labels) are given, and the goal is to find groups → unsupervised clustering.

03

Deep learning and generative AI

Learned features, why now, and models that create

Deep learning learns its own features

Classic machine learning



→ **Engineer designs features** dark-pixel fraction, edge length → **Simple model** e.g., a line or a decision rule → **crack / no crack**

Deep learning



→ **Layer 1** edges → **Layer 2** textures → **Layer 3+** crack-like shapes → **crack / no crack**

Why now?

Deep learning took off when three things came together

≈1.28 M

Data: labeled training images in the 2012 ImageNet challenge (1,000 categories)

2 GPUs

Compute: graphics cards trained the 2012 AlexNet image model in about a week

2017

Algorithms: the transformer design appears — the basis of today's large language models

Sources: Russakovsky et al., IJCV 2015; Krizhevsky, Sutskever & Hinton, NeurIPS 2012; Vaswani et al., NeurIPS 2017

Generative AI creates new content instead of a label or a number

Predictive AI

- Input → a number or a category
- Strength of this mix: 42 MPa
- This photo: crack
- Checked against a known answer

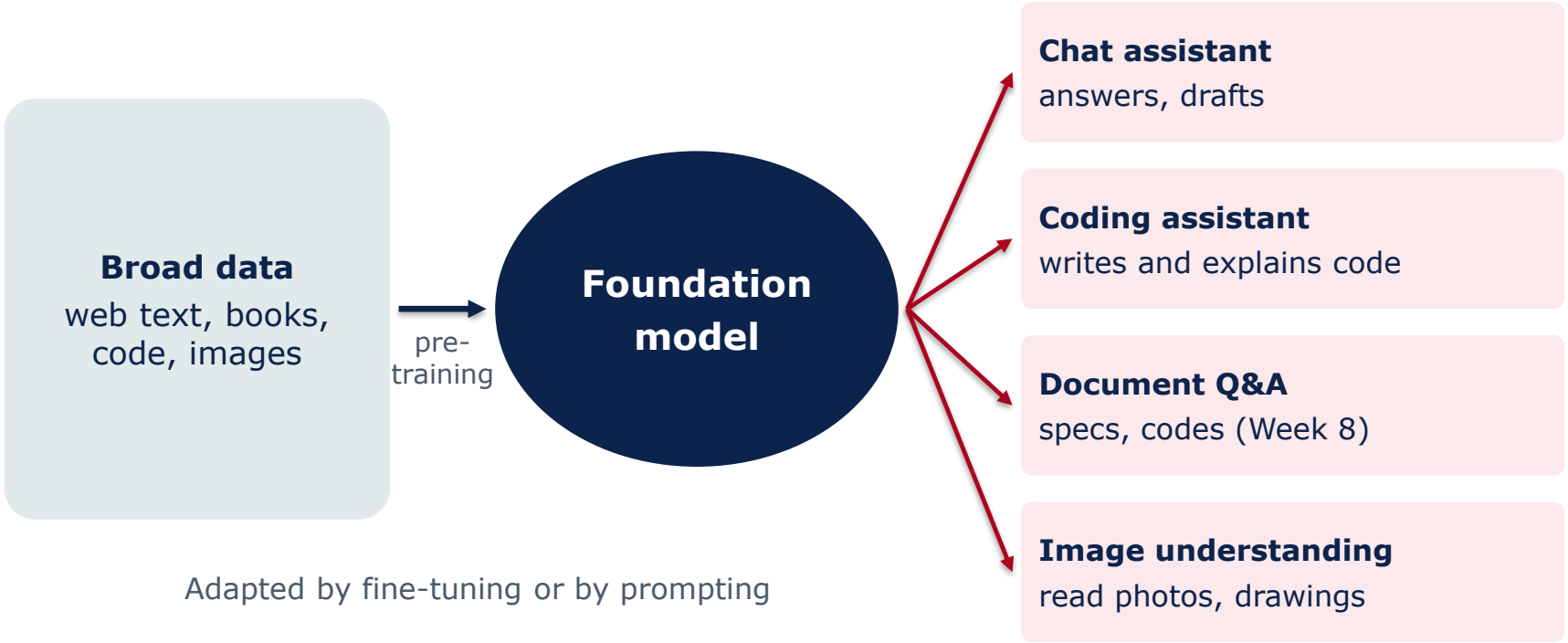
vs

Generative AI

- Prompt → new text, images, code, or designs
- Draft a method statement for a pour
- Concept images of a shaded façade
- Python code to plot sensor data

Generated content is **plausible**, not verified — an engineer must check it

Foundation models: train once on broad data, adapt to many tasks



Same idea, different inputs and outputs: eight CE and ARCE tasks

Task	Input	Output	Type of learning or task
Predict concrete strength	Mix proportions, age	Strength (MPa)	Supervised · regression
Detect cracks in photos	Photo (pixels)	Crack / no crack	Supervised · classification
Estimate a design's heating load	Shape, glazing, height	kWh/m² per year	Supervised · regression
Group buildings by energy use	Hourly meter data	Group number	Unsupervised · clustering
Flag unusual bridge readings	Strain, vibration data	Normal / unusual	Unsupervised · anomalies
Time traffic signals	Queue lengths	Green time per phase	Reinforcement learning
Draft a daily site report	Notes, photos	Report text	Generative AI
Make façade concept images	Text prompt	Images	Generative AI

Activity

Sorting activity: what kind of learning or task?

1 Estimate footing settlement (mm) from soil test results

2 Label drone photos of roofs "hail damage" or "no damage"

3 Find groups of similar trips in a year of unlabeled bike-share data

4 A crane controller improves load placement by trial and error in a simulator

5 Write a first-draft specification section from an engineer's bullet notes

6 Predict tomorrow's city water demand (m^3/day) from the weather forecast

7 Spot water-main pressure readings that look unlike any normal day

8 Predict whether a bridge deck will be rated "poor" at the next inspection

4 min · pairs

A · Supervised · regression

B · Supervised · classification

C · Unsupervised

D · Reinforcement learning

E · Generative AI (task)

Activity

Sorting activity: answers

#	Answer	The clue
1	A · regression	Output is a number (mm), learned from past cases with measured settlement
2	B · classification	Output is one of two labels
3	C · unsupervised	No labels; the goal is to find groups (clustering)
4	D · reinforcement	Learns from trial, error, and a reward
5	E · generative	Creates new text
6	A · regression	Output is a number (m^3/day)
7	C · unsupervised	No labels; find the unusual (anomaly detection)
8	B · classification	Output is a category (poor / not poor) — you will do this in Lab 11

04

Data is a design material

Quality, quantity, and representativeness

Data is a design material

Check your data the way you check your concrete

1 Quality

Errors, wrong units, missing values, duplicate rows. Garbage in, garbage out.

2 Quantity

Enough examples to cover the variety you care about — including rare cases like failures.

3 Representativeness

Does the data look like where the model will be used? Climate, materials, sensors, neighborhoods.

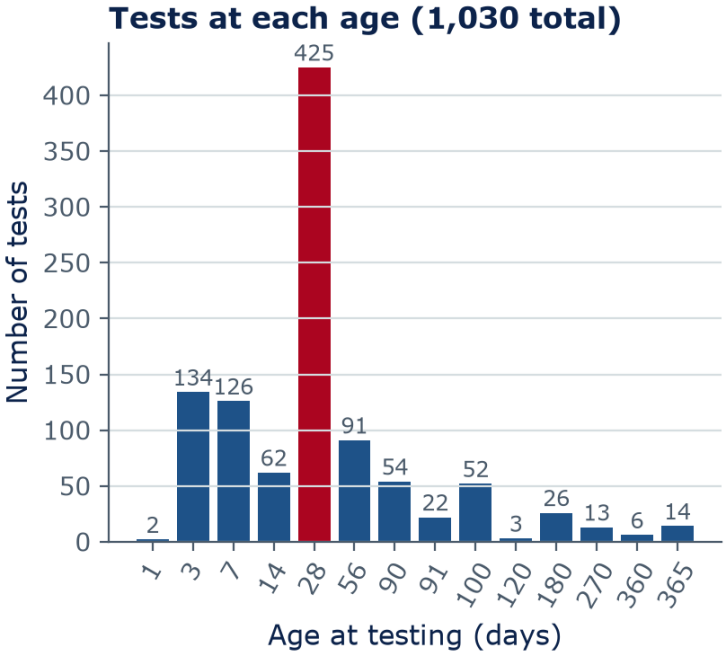
4 Labels

Who labeled it, how consistently? Two inspectors may rate the same deck differently.

Data is a design material

A model only knows the range of data it has seen

- Our concrete data: **425** of 1,030 tests are at 28 days, but only **14** at 365 days
- Cement ranges from 102 to 540 kg/m³ — outside that range, a model is **guessing**
- Laboratory mixes compiled for a 1998 study: field concrete (placement, curing, weather) can differ
- Before trusting a model, ask: **what data, from where, covering what range?**



Data: Yeh (1998), UCI concrete dataset

Concept check

Will the crack detector still work?

A crack detector scores very well on test photos of clean, dry campus walls. You want to use it on wet, shadowed bridge piers. What is the main concern?

A It has too many parameters to run on a laptop

B The new photos may not look like the training photos, so performance may drop

C A high test score guarantees it works on any concrete

D Crack detection should use reinforcement learning

Concept check

Will the crack detector still work? — answer

A crack detector scores very well on test photos of clean, dry campus walls. You want to use it on wet, shadowed bridge piers. What is the main concern?

A It has too many parameters to run on a laptop

B The new photos may not look like the training photos, so performance may drop

C A high test score guarantees it works on any concrete

D Crack detection should use reinforcement learning

Representativeness: test on photos like the ones you will really use before trusting it.

Lab 02

Lab day: working with real engineering data

LAB

02

Bring a charged laptop

Open the notebook from the course GitHub page

- Load `concrete_strength.csv` with **pandas**: `head`, `shape`, `describe`
- Select, filter, sort, and group the tests by age; add a water–cement ratio column
- Plot with **matplotlib**: a histogram and scatter plots of strength vs cement, age, and w/c
- Name X and y, spot outliers, write short observations
- **Exit ticket:** 3 short questions you answer alone, without AI, at the end of lab

Homework 1

Homework 1 is posted: AI or not? Exploring engineering data

Due **end of Week 3** — Sunday 11:59 pm on Brightspace. Handout on GitHub.

AI-FREE

Part A

Place 8 systems on the autonomy–adaptivity map, with reasons

AI-FREE

Part B

Classify 6 tasks as supervised, unsupervised, or reinforcement learning

AI-ASSISTED

Part C

Explore the building energy dataset with pandas and plots

AI-FREE

Part D

A short reflection on AI in your own studies or future work

Wrap-up

Key takeaways

1. Machine learning flips programming: give it **data + answers**, and it produces the rules (a model)
2. **Features X** go in, the **target y** comes out; one row of data = one example
3. Supervised (regression, classification) · unsupervised (groups, anomalies) · reinforcement (rewards)
4. Deep learning learns its own features; generative AI creates content; foundation models are reused
5. A model is only as good as its data: quality, quantity, labels, and representativeness

Learn more

Readings and resources

- **Elements of AI, Chapter 4, Section I: The types of machine learning** — about 20 min
- **Google: Introduction to Machine Learning** — what ML is and its types, about 20 min
- **3Blue1Brown: But what is a neural network?** — video, 19 min — a preview of Week 5
- **Course GitHub: Week 2 README** — key terms and check-yourself questions